

Assessing Bicycle Lane Quality Using Smartphone Sensor Data

5ARE0 – Assignment 1

Group 41

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1 Business and Objective Understanding

Cycling is only as safe and comfortable as the lane it happens on. Cracks, potholes, root bumps and raised joints make a ride tiring, they push the rider into sudden corrections, and they increase the risk of falls. At the same time, cities cannot afford to inspect every kilometre of bike lane by hand. Everyone who rides already carries a phone with an accelerometer, a gyroscope and a gravity sensor, so the idea is to let the rides themselves do the inspection.

Our objective is a binary classifier that, given a short time window of phone sensor data recorded on a bike, says whether the lane under the wheels is *smooth* or *bumpy*. From the health and monitoring side, this is a cheap proxy for the vibration a rider is exposed to and for the places where a lane is likely to cause discomfort or accidents. From the city side, it is a way to rank lanes for maintenance.

We define success as building a robust model that accurately detects bumpy roads to guide maintenance interventions, while remaining computationally lightweight for smartphone deployment and generalizing well to unseen data.

2 Data Understanding

2.1 Collection methodology

We recorded with the free *Sensor Logger* app, with only the accelerometer, gyroscope and gravity sensors enabled, the default 100 Hz sampling rate. A recording starts with the screen tap on the phone, the rider cycles along one lane, and the recording stops with another tap.

Placement. The assignment suggests a pocket, an armband or a waistband. We chose none of them. We fixed the phone *vertically on the straight part of the handlebar*, screen facing the rider, using a bike phone support. The reasoning is that a pocket or a waistband mixes the road signal with leg movement and pedalling, and the phone moves relative to the body. A rigid handlebar mount sees the road through the front wheel and the fork, and it moves with the bike, so what is left is mostly the motion of the bike on the surface. The price is that the support is not identical across bikes: the tilt of the holder changes from one bike (and one session) to another, which is why we need the alignment step in Section 3.1.

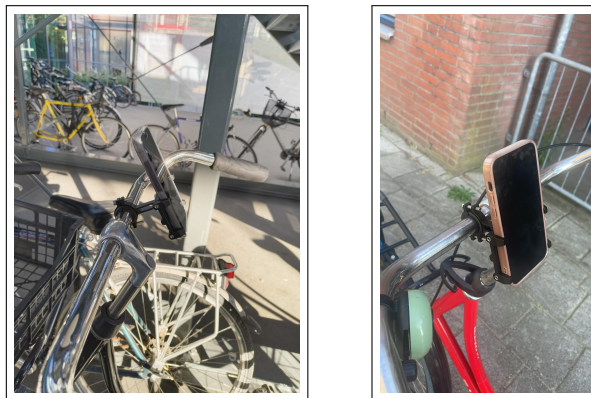


Figure 1: Phone mounting setup. The phone is fixed vertically on the straight part of the handlebar using a rigid bike phone support. The two images show the mount from different angles and on different bikes, highlighting the varying tilt that necessitates the gravity-based alignment step.

Dataset. We have 27 (13 bumpy and 14 smooth) recordings in and around Eindhoven (campus roads, the station area, the stadium area, residential streets), collected in September 2026 by 3 riders on 3 iPhones. Each recording was labelled *at recording time* as smooth or bumpy by the rider: smooth means a regular red cycling lane without imperfections, bumpy means a lane with visible rough features (potholes, cracks, raised joints, uneven surface) that is still safe to ride. The label lives in the recording name.

2.2 Exploratory analysis

Mounting differs between recordings. The phone holder orientation varies substantially across rides, with estimated pitch angles ranging from about 0° to -40° . Without correcting for these differences, the same road vibration could appear on different sensor axes, causing features to reflect the mounting setup rather than the road surface. This motivates the gravity-based alignment step described in Section 3.1.

After alignment, the residual error was below 0.05° for all recordings, indicating highly consistent axis alignment.

Frequency content. Before designing the filter, we analysed the Welch power spectral density (PSD) and spectrograms of the aligned signals.:

- *Accelerometer.* All three axes (X, Y, Z) show a broad hump in the 10–20 Hz range, likely corresponding to rolling, pedalling, and frame resonance. The separation between classes is not defined by a single sharp peak, but by the overall amplitude: above roughly 20 Hz, the bumpy PSD is consistently higher than the smooth PSD on all three axes.
- *Gyroscope.* The separation is clearest on the X axis, which shows elevated energy from about 5 Hz up to 30 Hz. On the Y and Z axes, the two classes are almost overlapping, with only a very slight elevation for bumpy roads. Below roughly 5 Hz the gyroscope is dominated by steering and leaning.
- *Spectrogram.* The bumpy spectrograms show a sustained elevated band instead of isolated vertical streaks (Figure 2). Our bumpy recordings therefore capture continuous roughness more than isolated potholes.
- *High-frequency noise.* Above ~ 40 Hz (close to Nyquist) the classes still diverge slightly, but we treat this as sensor noise and leave it out of the band.

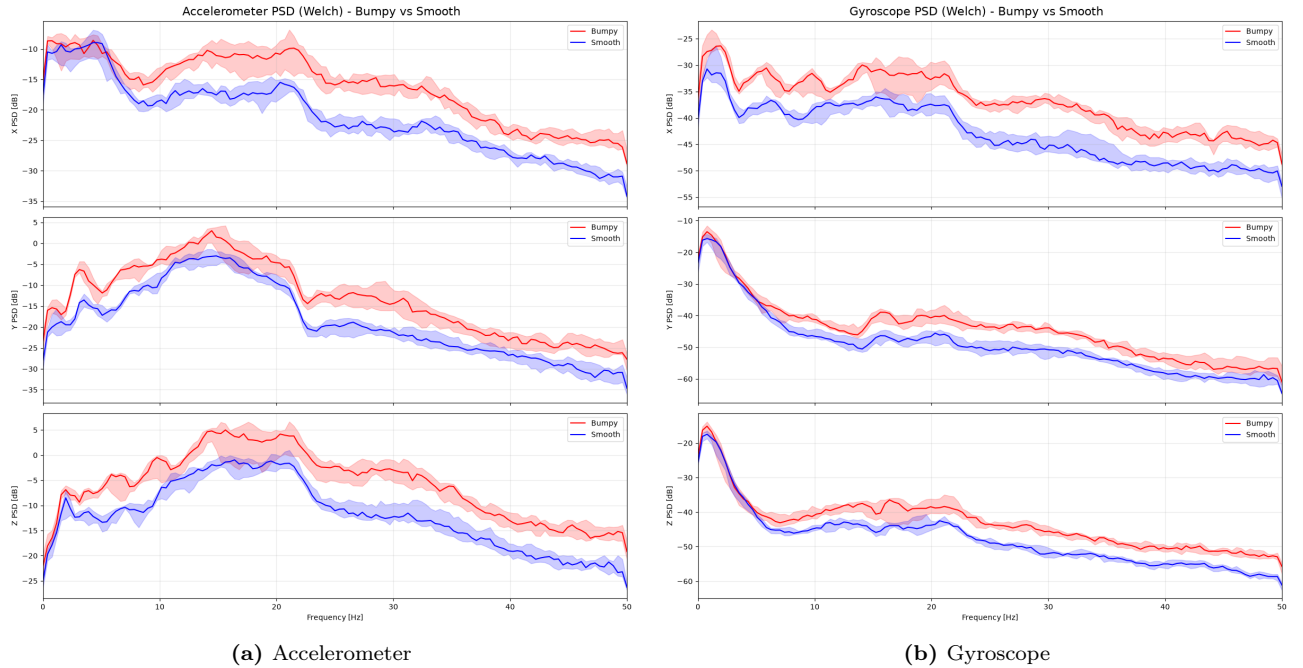


Figure 1: Welch PSD of the aligned signals, median and inter-quartile range over the bumpy (red) and smooth (blue) recordings.

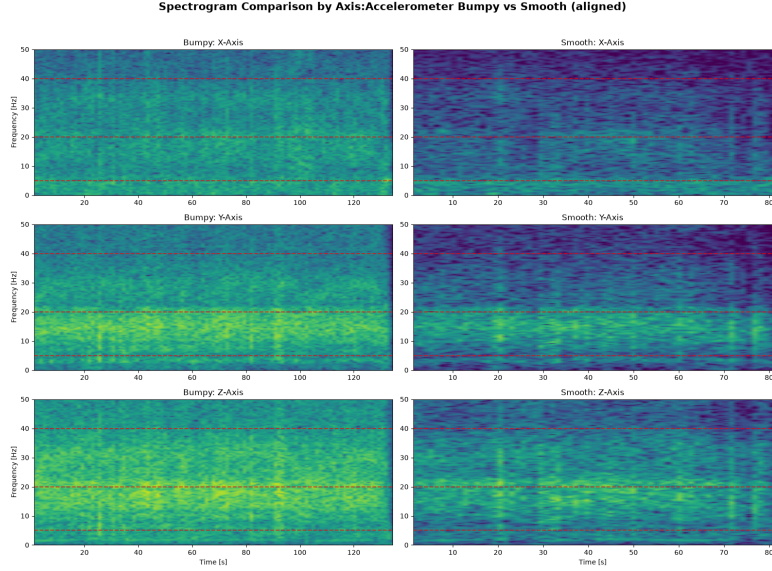


Figure 2: Accelerometer spectrograms (aligned) of one bumpy and one smooth recording on the same axes. Dashed lines mark 5, 20 and 40 Hz.

2.3 Data quality and limitations

- **Sensor synchronisation.** The three sensors in a single recording can be slightly out of sync, differing by at most 5 samples in length. We simply truncate all three to the shortest one.
- **Alignment quality.** The gravity-based alignment is highly stable. After rotation, the median gravity vector is within 0.04° of the target $(0, -g, 0)$ for every recording. Furthermore, the phone does not drift in its holder: the angle between the median gravity of the first and last 20% of each ride is at most 1.98° .
- **Small, correlated dataset.** To be realistic, 27 recordings is a small dataset. Because we use 3-second overlapping windows, most of our 1047 total windows may be correlated. They are not 1047 independent samples. This is exactly why every train/test split in this work is strictly grouped by recording ID; a random split would be pure data leakage.
- **Class and session confounding.** This is our biggest physical limitation. The mounting pitch tends to cluster by session. For example, we have a cluster of rides at $\approx -25^\circ$ pitch that are mostly smooth, and another at $\approx -35^\circ$ that is a mix. A classifier could easily “cheat” by learning the specific vibration signature of a specific bike, phone, or tyre, rather than the actual road texture. Because we group by recording, we cannot fully decouple this effect. Additionally, we did not record speed, which directly scales the vibration amplitude.
- **Label noise.** We labeled these recordings ourselves based on physical feel. A recording labeled “smooth” might still contain a few small bumps or raised manhole covers, but every single 3-second window inside it inherits the “smooth” tag.
- **No GPS.** Without GPS data, we cannot correct for road slope or steering angle, and we cannot map our predictions back to a physical location on a map.

3 Data Preparation

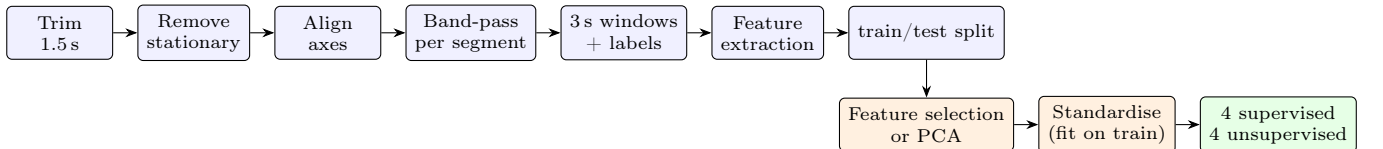


Figure 3: Blue steps are applied before model training and use no information from the test recordings. Feature selection, PCA and standardisation (orange) are fitted using training data only to prevent data leakage.

3.1 Preprocessing pipeline and rationale

1. **Sign convention.** iOS and Android report accelerometer and gravity with opposite signs (inertial vs. applied force); the gyroscope has the same sign on both. We convert everything to the iOS convention, using

the recording metadata (platform and standardisation flag) and, if that is missing, the sign of the gravity y component. With this convention a bike at rest on flat ground has gravity $\approx (0, -9.81, 0)$.

2. Trimming. The first and last 1.5 s (150 samples) of every recording are cut. They contain the taps on the screen.

3. Stationary segments. A stop at a traffic light says nothing about the lane. We compute the squared accelerometer magnitude, average it over a rolling window of 50 samples (0.5 s), and mark a sample as stationary if this energy is below $0.30 \text{ (m/s}^2\text{)}^2$. Only runs of at least 200 samples (2 s) are removed, so short pauses in a ride survive. The removal leaves time gaps, so later steps that depend on continuity (filtering, windowing) are done per continuous segment.

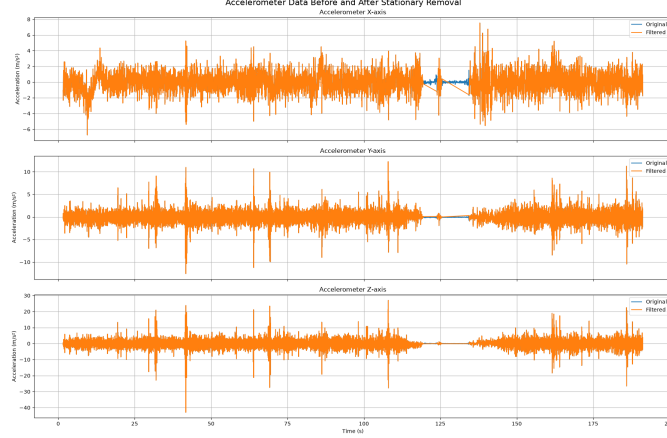


Figure 4: Accelerometer of `smooth_con_fermata` before (blue) and after (orange) stationary removal. The straight line in the X-axis is only the plot joining the two sides of the removed stop.

4. Axis alignment. We want every recording expressed in the same *bike frame*: x along the handlebar towards the rider's left, y perpendicular to the road (down), z in the travel direction. The holder has two degrees of freedom, a roll ψ of the phone around its screen normal and a pitch θ of the holder around the handlebar. With $u = -g_{\text{med}}/\|g_{\text{med}}\|$ the unit vector pointing up, in the phone axes :

$$u = (\cos \theta \sin \psi, \cos \theta \cos \psi, -\sin \theta), \quad \psi = \text{atan2}(u_x, u_y), \quad \theta = \text{atan2}\left(-u_z, \sqrt{u_x^2 + u_y^2}\right). \quad (1)$$

The resulting rotation is applied to the accelerometer, gyroscope and gravity signals of the recording, bringing all rides into a common bike-centered reference frame.

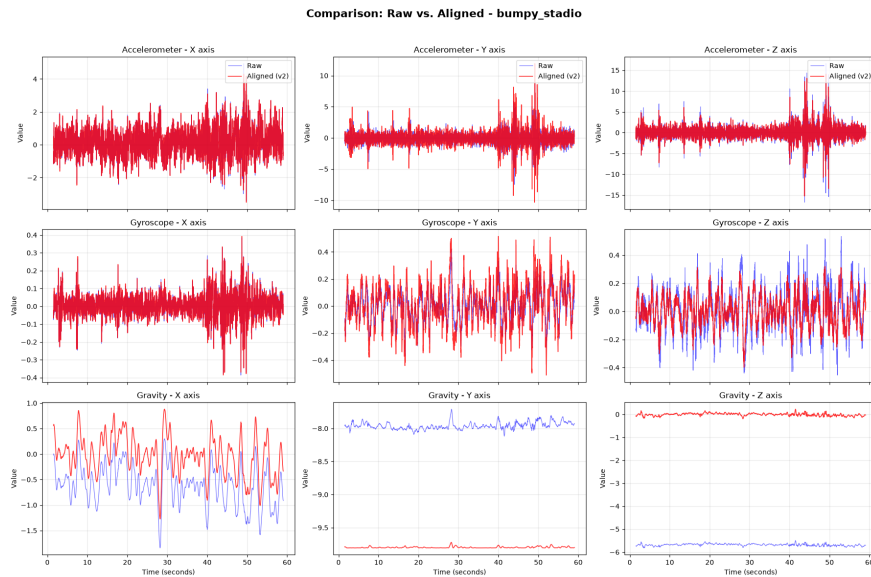


Figure 5: `bumpy_studio`, raw (blue) vs. aligned (red). Accelerometer and gyroscope keep their dynamics but move between axes; the gravity components are what the alignment is built on (the aligned gravity is a constant $(0, -9.8, 0)$).

5. Band-pass filter. Based on the spectral analysis, we apply a 4th-order Butterworth band-pass filter using `filtfilt` (zero-phase filtering). The passband is 10–40 Hz for the accelerometer and 5–40 Hz for the gyroscope.

The upper cutoff removes frequencies close to Nyquist, where sensor noise is more likely, while the lower cutoff removes gravity residuals, pedalling effects, and low-frequency steering or leaning motions.

The PSDs show that bumpy roads are not characterised by a single dominant frequency but by consistently higher energy across a broad frequency range. In particular, the gyroscope X-axis separates the two classes from around 5 Hz onward, while the accelerometer differences are most evident above 10 Hz. Filtering is performed separately on each continuous riding segment. Gaps created by stationary-segment removal are not filtered across, as the discontinuities would introduce artificial broadband energy. A new segment is started after gaps longer than 0.2 s, and segments shorter than 0.5 s are discarded.

The frequency bands were chosen after visual inspection of the spectra from all recordings. Although this introduces a small risk of information leakage, the bands are broad and only guide preprocessing rather than model fitting, so we expect the effect to be minimal.

6. Windowing and labels. We cut 3 s windows with 30% overlap, only inside a continuous segment, and each window inherits the label of its recording (bumpy = 1, smooth = 0). Three seconds contain many periods even of the lowest passband frequency and give a Welch resolution of 0.67 Hz, enough to resolve bands of 5–10 Hz; overlap increases the number of windows and prevents a single bump from being cut in half by a window boundary. We chose these values a priori and did not run a sensitivity study on the window length. The result is **1047 windows**, each carrying its recording as a group identifier.

3.2 Feature engineering, extraction

Each window is described by features computed on each of the six channels (accelerometer and gyroscope, X/Y/Z), always on the band-passed signal. Time-domain features capture vibration intensity and extreme impacts, while frequency-domain features capture how vibration energy is distributed across frequencies.

- **Time domain (8):** mean, standard deviation, RMS, maximum, minimum, peak-to-peak, energy, and peak count.
- **Frequency domain (6):** dominant frequency, spectral centroid, spectral entropy, spectral flatness, geometric mean and arithmetic mean of the normalised PSD (Welch with 150-sample segments, restricted to the passband of the sensor).
- **Band energies:** accelerometer 10–20, 20–30, 30–40 Hz; gyroscope 5–10, 10–20, 20–30, 30–40 Hz.

This produces **105 features** per window, but many of them overlap in the information they capture. For example, after band-pass filtering, features such as standard deviation, RMS, and energy become strongly correlated. Instead of manually deciding which features to keep, we rely on the feature-selection step to remove redundant and uninformative features using mutual information and correlation analysis.

3.3 Dataset Splitting

3.3.1 Train / Test Split

Before any feature selection, the dataset is split **by recording**: all the windows of a recording go to the same set, so the model is never tested on a recording it has already seen.

- **Test set** (15% of the recordings of each class): kept aside and used only for the final evaluation of the models.
- **Training set** (the remaining recordings): the same set is used by both the supervised and the unsupervised feature selection.

The same share of recordings is taken from each class, so both bumpy and smooth recordings are present in the test set.

3.4 Feature Selection

Selection is different for supervised and unsupervised models. This is intentional. The supervised pipeline uses the class labels to identify informative features, whereas the unsupervised models must not have access to any label information. Reusing the supervised feature set for clustering would introduce *target leakage*, since the clustering algorithms would indirectly benefit from features chosen specifically to separate the true classes, leading to overly optimistic clustering results.

- **Supervised** (one feature set per model, computed only on the training set):
 - (i) The training set is further split into an **inner train** and a **validation** set (20% of the remaining recordings).
 - (ii) Features are ranked by mutual information with the label, computed on the inner train only.

- (iii) A correlation filter removes features with $|\rho| > 0.8$ to an already selected feature.
- (iv) The remaining features are evaluated greedily in MI order. A feature is kept if fewer than four features have been selected or if it improves the validation macro-F1 by more than 0.01. Selection stops at 12 features.
- **Unsupervised** (no labels used at any stage):
 - (i) Low-variance features and highly correlated features ($|r| > 0.8$) are removed.
 - (ii) The remaining features are standardized and projected using PCA. Up to 20 principal components are retained (with at least 5 evaluated).
 - (iii) Four clustering methods are evaluated: K-means, Fuzzy C-means, Gustafson-Kessel, and Hierarchical clustering.
 - (iv) For each method, the number of PCA components is chosen by minimizing the combined rank of the Gath-Geva and Xie-Beni validity indices.

All feature selection, scaling, and PCA fitting are performed exclusively on the training data, ensuring that the test set remains completely unseen until the final evaluation.

Normalisation. Features are standardised (z -score) using a `StandardScaler` fitted on the training data and applied unchanged to the validation and test sets. This is particularly important for KNN, SVM, and Logistic Regression, which are sensitive to feature scale, and for PCA, where high-variance features would otherwise dominate the principal components. Random Forest is largely unaffected by feature scaling.

4 Modeling

Validation scheme. To prevent data leakage, the dataset is split at the recording level, so all windows from a recording belong to the same set. Approximately 15% of the recordings from each class are reserved for testing, while the remaining recordings form the training set. For supervised feature selection, the training set is further divided into an inner training set and a validation set. The test recordings remain completely unseen until the final evaluation.

Supervised models. We selected four classifiers with different modelling assumptions: **K-Nearest Neighbours** ($k = 5$, local and non-parametric), **Logistic Regression** (linear and interpretable), **Random Forest** (non-linear and robust to feature scale), and **Support Vector Machine** with an RBF kernel. Each model receives its own feature subset obtained through the supervised feature-selection procedure described above.

Unsupervised models. We evaluate four clustering approaches: **K-Means**, **Fuzzy C-Means (FCM)**, **Gustafson-Kessel (GK)**, and **Hierarchical clustering**. Prior to clustering, the selected features are standardised and projected into a lower-dimensional representation using PCA. Since no labels are used during training, clustering quality is assessed only afterwards by comparing the resulting clusters with the true road-surface labels.

Comparison methodology. All models are trained and evaluated using the same recording-level train/test split (85% train, 15% test), keeping the test set completely unseen during feature selection, PCA fitting, and training. Supervised models are evaluated using accuracy, macro-F1, and per-class recall, with macro-F1 as the primary metric. For unsupervised models, clustering quality is assessed via the Adjusted Rand Index (ARI); to enable direct comparison, clusters are mapped to their majority class to compute accuracy and macro-F1. Given the limited number of independent recordings (27), small performance differences should be interpreted cautiously.

5 Evaluation

5.1 Performance metrics and results

Figure 6 summarises the evaluation results for all supervised and unsupervised models on the held-out test set. The reported metrics include train and test accuracy, balanced accuracy, macro precision, macro recall, macro F1-score, Adjusted Rand Index (ARI), Normalised Mutual Information (NMI), and the majority-class baseline. Together, these metrics provide a comprehensive view of both predictive performance and cluster quality.

Among the supervised models, Logistic Regression achieved the highest test accuracy (86.9%) and macro F1-score (0.869), closely followed by SVM (86.3%) and KNN (85.7%). Random Forest reached perfect training accuracy but lower test performance (82.1%), suggesting some degree of overfitting. The relatively small perfor-

Model	Train accuracy	Test accuracy	Train balanced accuracy	Test balanced accuracy	Test macro precision	Test macro recall	Test macro F1	Train ARI	Test ARI	Train NMI	Test NMI	Majority baseline
KNN	0.944	0.857	0.941	0.858	0.862	0.858	0.857	0.789	0.507	0.686	0.422	0.512
Logistic Regression	0.893	0.869	0.883	0.871	0.876	0.871	0.869	0.617	0.542	0.511	0.463	0.512
Random Forest	1.000	0.821	1.000	0.823	0.830	0.823	0.821	1.000	0.410	1.000	0.343	0.512
SVM	0.920	0.863	0.909	0.865	0.874	0.865	0.863	0.706	0.525	0.619	0.459	0.512
K-means	0.856	0.756	0.838	0.761	0.824	0.761	0.745	0.504	0.258	0.419	0.322	0.512
Gustafson-Kessel	0.895	0.815	0.885	0.819	0.845	0.819	0.813	0.624	0.395	0.520	0.379	0.512
Fuzzy c-means	0.869	0.780	0.861	0.783	0.802	0.783	0.777	0.544	0.309	0.435	0.278	0.512
Hierarchical	0.851	0.738	0.834	0.744	0.825	0.744	0.722	0.491	0.223	0.404	0.331	0.512

Figure 6: Summary of the evaluation metrics for all supervised and unsupervised models. Metrics include train and test accuracy, balanced accuracy, precision, recall, F1-score, ARI, NMI, and the majority-class baseline.

mance differences between the supervised models indicate that the quality of the extracted features contributes more to performance than the specific classifier.

The unsupervised models showed greater variability. Gustafson-Kessel achieved the highest test accuracy (81.5%), the highest macro F1-score (0.813), the highest test ARI (0.395), and the highest test NMI (0.379). K-Means, Fuzzy C-Means and Hierarchical clustering produced progressively lower scores across most metrics, with Hierarchical performing worst (test accuracy 73.8%, test ARI 0.223). Although several clustering approaches achieved encouraging performance, these results should be interpreted cautiously, as some of the discovered structure may be related to rider, bike, or session characteristics rather than road quality alone.

5.2 Selected features

The selected feature subsets typically contained between 4 and 12 features, depending on the model. The most consistently selected features were gyroscope X-axis band energy (20–30 Hz), accelerometer Y-axis band energies (20–30 Hz and 30–40 Hz), and accelerometer X-axis standard deviation.

These features are physically meaningful. Band-energy measures quantify how much vibration energy is present in specific frequency ranges, while standard deviation captures overall vibration intensity.

Their frequent selection is consistent with the PSD analysis, which showed the clearest differences between smooth and bumpy roads in the 20–40 Hz region. More generally, the selected subsets were dominated by vibration-energy features rather than raw statistical descriptors, suggesting that frequency content carries more discriminative information than individual signal amplitudes alone.

5.3 Model comparison, justification and error analysis

Table 1: Summary of the evaluated models.

Model	Main strengths	Main limitations
KNN	Simple, non-parametric, captures local patterns	Sensitive to feature scaling and noisy samples
Logistic Regression	Interpretable, fast, works well on small datasets	Assumes a linear decision boundary
Random Forest	Captures non-linear relationships, robust to irrelevant features	Less interpretable, may overfit small datasets
SVM (RBF)	Flexible non-linear boundary, effective with few features	Limited interpretability, sensitive to parameter choices
K-Means	Simple and computationally efficient	Assumes roughly spherical clusters
Fuzzy C-Means	Allows partial membership to multiple clusters	Sensitive to initialization and noise
Gustafson-Kessel	Can model clusters with different shapes and orientations	More computationally complex
Hierarchical	Does not require explicit cluster centres	Sensitive to linkage choices and noise

Error analysis. Although the overall performance is encouraging, several sources of error are expected. First, some recordings labelled as smooth contain occasional raised joints or small irregularities, introducing label noise.

Second, manoeuvres such as braking, cornering, and acceleration can generate vibration patterns similar to those produced by rough road surfaces. Third, recordings collected on bikes, phones, or mounting configurations that are underrepresented in the training set may be more difficult to classify correctly. Finally, the small number of independent recordings limits the ability of the models to generalise to unseen riding conditions.

6 Deployment Considerations and Recommendations

Deployment on the external dataset. After model selection, Logistic Regression, which was found to be the best model, is retrained on all available recordings and deployed on the independent external dataset provided for the assignment. The external recordings are never used during model development, feature selection, parameter tuning, or evaluation. To ensure consistency, the external dataset passes through exactly the same pipeline used for our recordings: sign-convention correction, trimming, stationary-segment removal, gravity-based axis alignment, band-pass filtering, windowing, and feature extraction. The same selected features are retained, and standardisation is performed using the mean and standard deviation previously learned from the training data. If dimensionality reduction is used, the PCA transformation learned during training is applied unchanged. Each 3-second window receives a smooth or bumpy prediction from the selected model. The resulting predictions are then visualised along the duration of the ride, highlighting which sections of the route are classified as smooth and which are classified as bumpy.

Practical implications. The pipeline is computationally lightweight. A window requires only filtering, feature extraction, and inference on a small number of features, making smartphone deployment feasible. The dominant source of latency is the 3-second window length itself.

Limitations.

- Only 27 independent recordings are available. Although windowing produces 1047 samples, the effective sample size is still determined by the number of recordings rather than the number of windows.
- Possible confounding of the class with bike, phone, rider, session, and speed (Section 2.3). The model may therefore partly learn properties of the recording setup rather than road quality alone.
- No GPS information was recorded, preventing speed normalisation, slope correction, and spatial mapping of predictions.
- Labels are assigned at the recording level rather than the segment level. Consequently, occasional bumps may appear within recordings labelled as smooth and vice versa.
- The filter bands, window size, and stationary-removal thresholds were chosen once and were not evaluated through a dedicated sensitivity analysis.

Recommendations and future work.

- Collect a larger and more diverse dataset, including the same roads ridden by different riders, bikes, and phones at multiple speeds.
- Evaluate generalisation using session-level or bike-level train/test splits to better isolate road-quality information from recording-specific effects.
- Log GPS information to enable speed-normalised features and spatial visualisation of road quality.
- Move from recording-level labels to segment-level labels and eventually to a continuous roughness score.
- Extend the system with dedicated pothole-event detection in addition to overall pavement-quality assessment.
- Investigate links between vibration-based features and cyclist comfort or vibration-exposure measures.

Use of Artificial Intelligence. Artificial intelligence tools were used in a limited supporting role for debugging assistance, brainstorming, language refinement, and proofreading of the report. AI was not used to generate the project code, develop the methodology, perform the analysis, select the models, or produce the reported results. All implementation, experimental design, data processing, model development, and final conclusions were carried out by the authors.